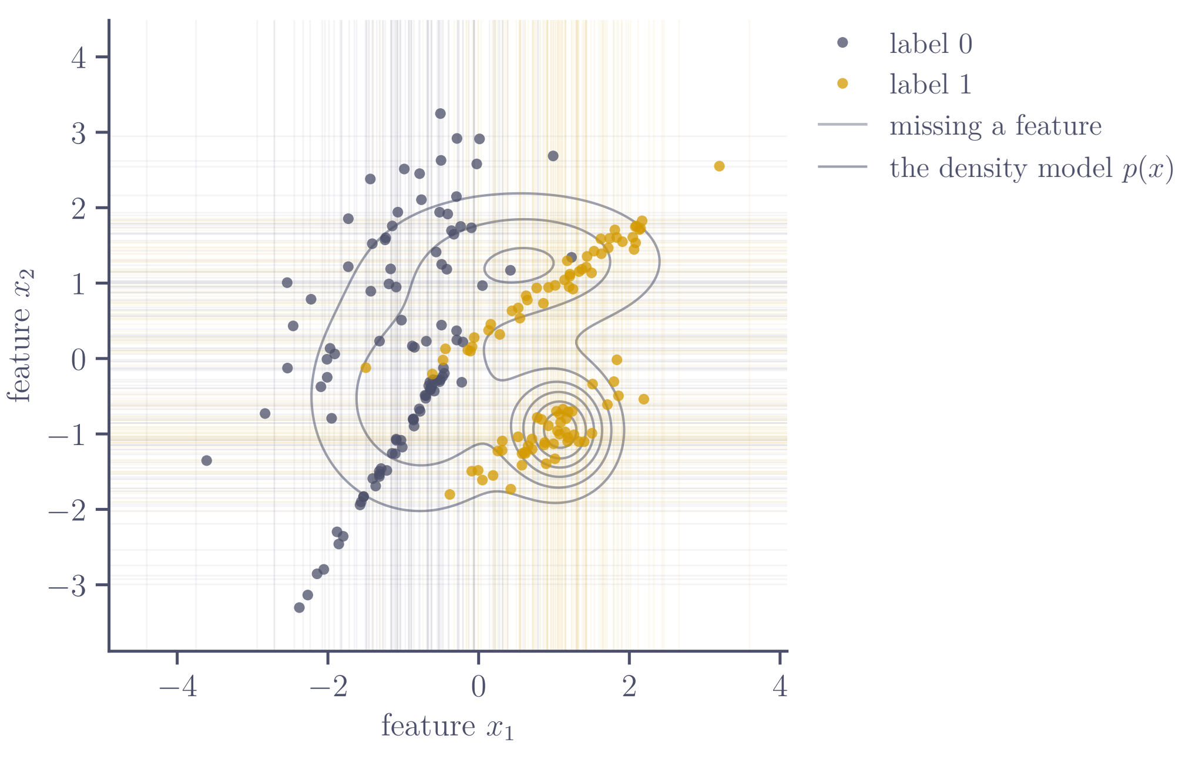
SETUP

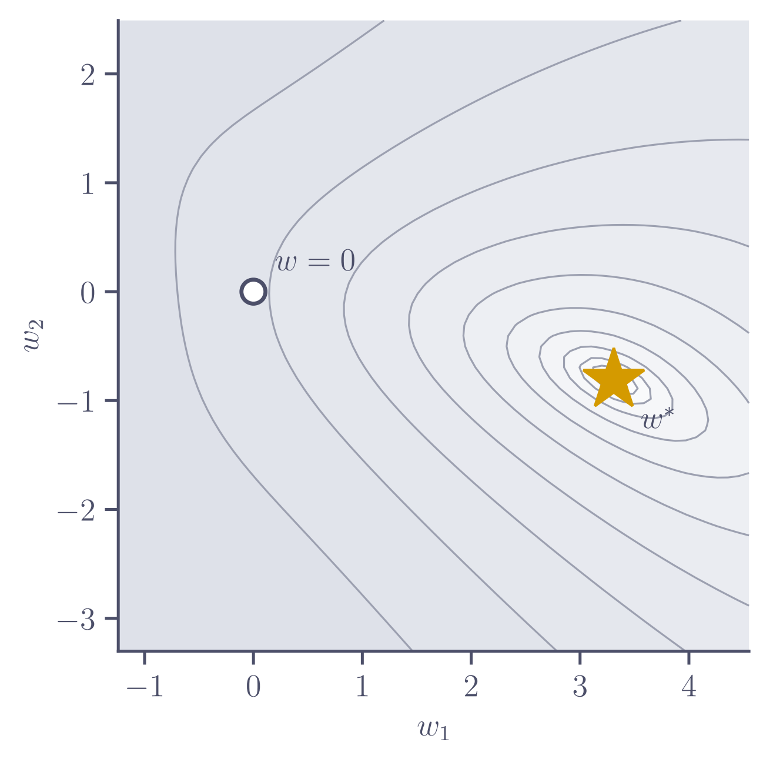
Training points with a missing feature



600 training points with two features, yellow label 1, slate label 0. 402 of them miss one feature. They are drawn as lines, because the missing value could be anywhere on them. A density model $p(x)$ (grey contours) is fit on the points first and then kept fixed. The classifier to train is a logistic regression $\sigma(w^\top x)$, the probability of label 1, with a boundary through the origin. Its direction and steepness are set by w, which training has to find.

GOAL

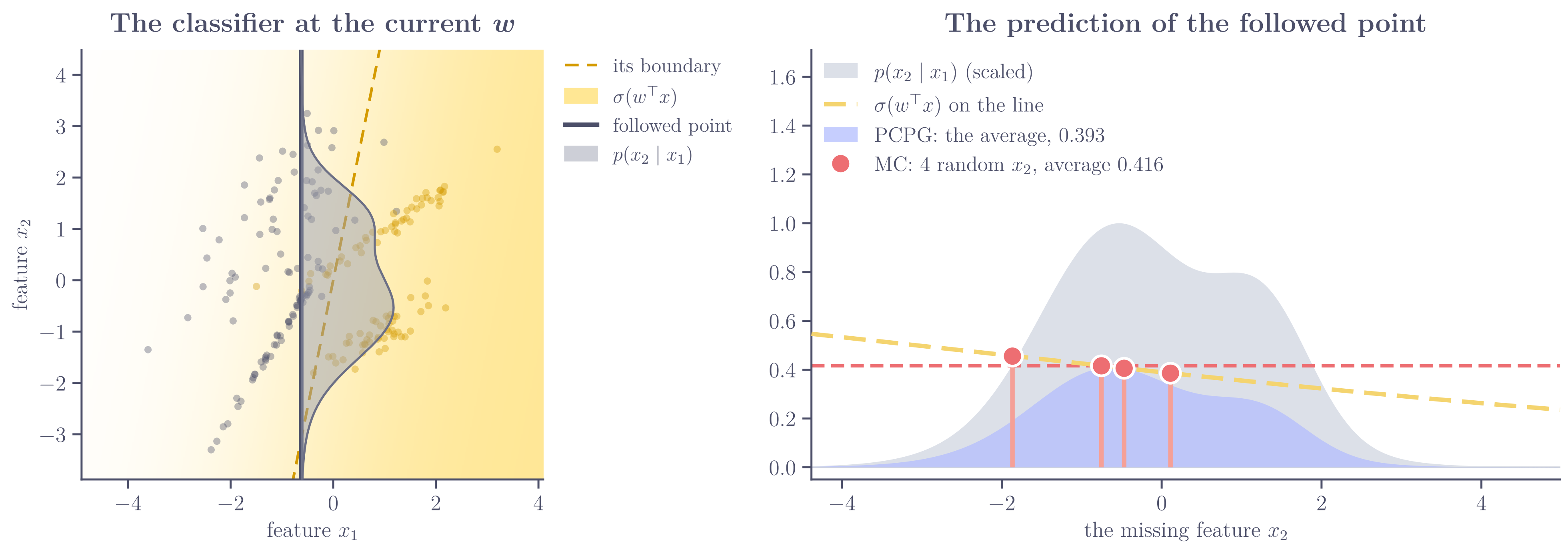
Find the w with the smallest loss $L(w) = \frac{1}{n}\Sigma_i - \log p(y_i \mid x_i^o)$



With two features, every w is a point in this plane, and every point is a classifier. The loss $L(w)$ says how badly that classifier predicts the training points: the average of $-\log p(y_i \mid x_i^o)$, which is small when every point gets a high probability of its own label. Grey is the loss for every w, lowest at w^* (star). Training starts at $w = 0$ and has to get to w^*. For that it needs the loss, and so the prediction of every point, again and again.

STEP 1

Predict every point with the current w



The classifier at the current w

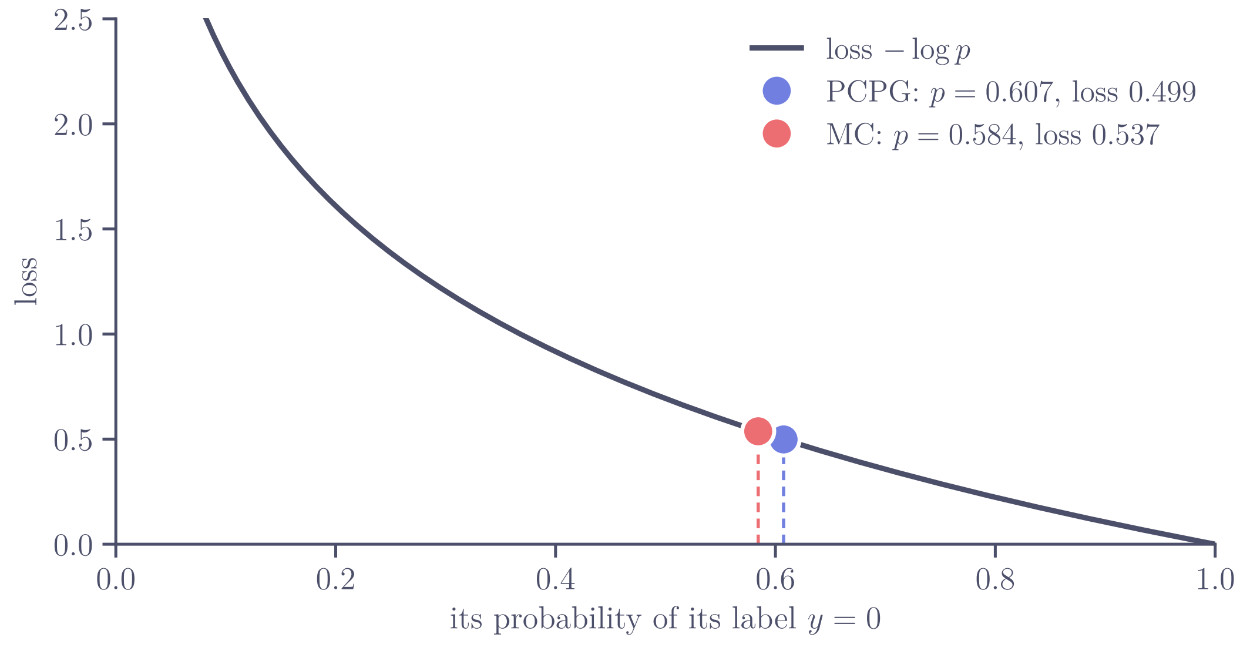
The prediction of the followed point

Training goes in steps. Here the current w is the one after 15 steps. Every point gets a prediction. For a complete point it is just $\sigma(w^\top x)$, the colour at the point. For an incomplete point, like the one followed here (line), it is the average of σ along its line, weighted by the density model $p(x_2 \mid x_1)$ (grey). PCPG computes this average exactly: 0.393. MC tries 4 random values of x_2 from the density model and averages σ at them: 0.416. Other random values would give a different number.

STEP 2

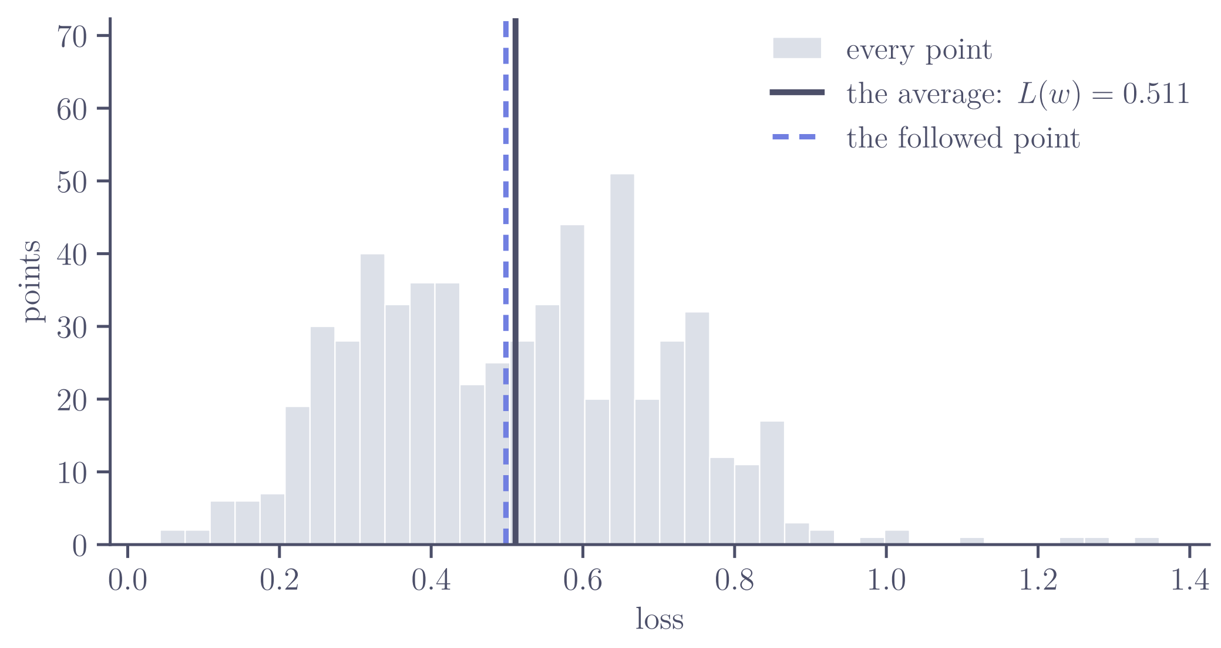
Measure how wrong the predictions are: the loss

The loss of the followed point



The followed point has label 0, so what counts is its probability of label 0: 0.607. Its loss is $-\log$ of that: small when the probability is close to 1, large when it is close to 0. With MC's estimate 0.584 the loss would be 0.537.

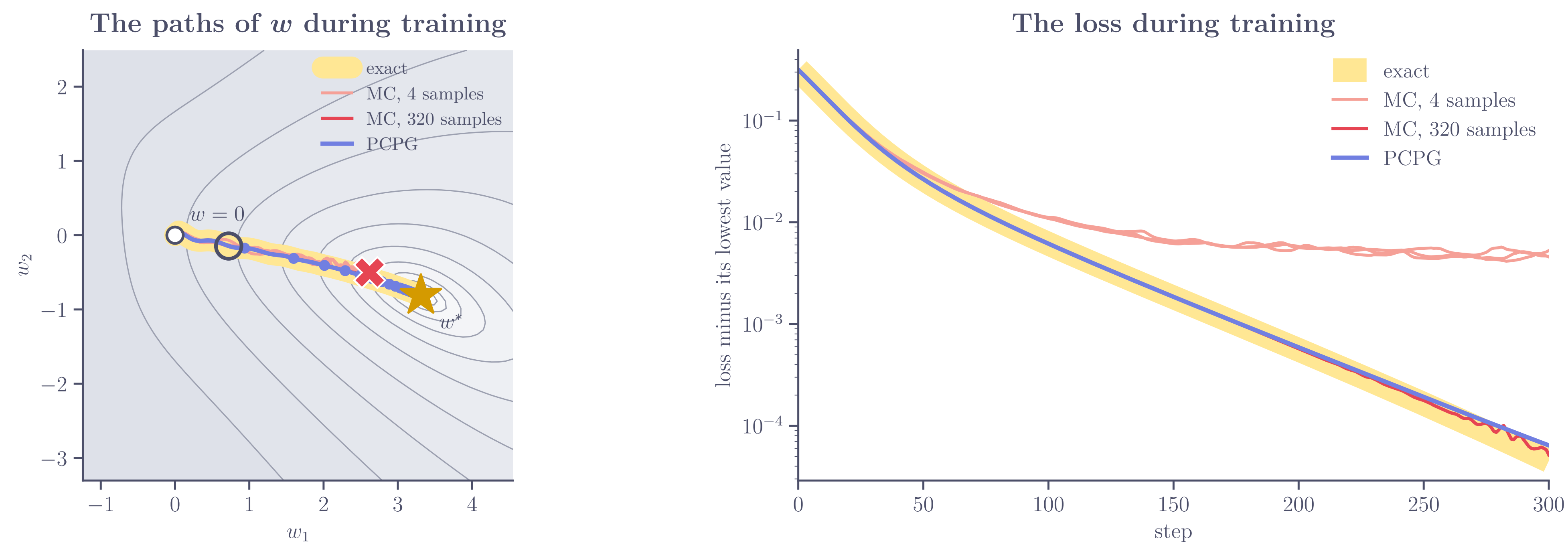
The losses of all points



The losses of all 600 points at the current w. Their average is the loss $L(w) = 0.511$. Training wants to make it smaller.

STEP 3

Change w a little to lower the loss, and repeat



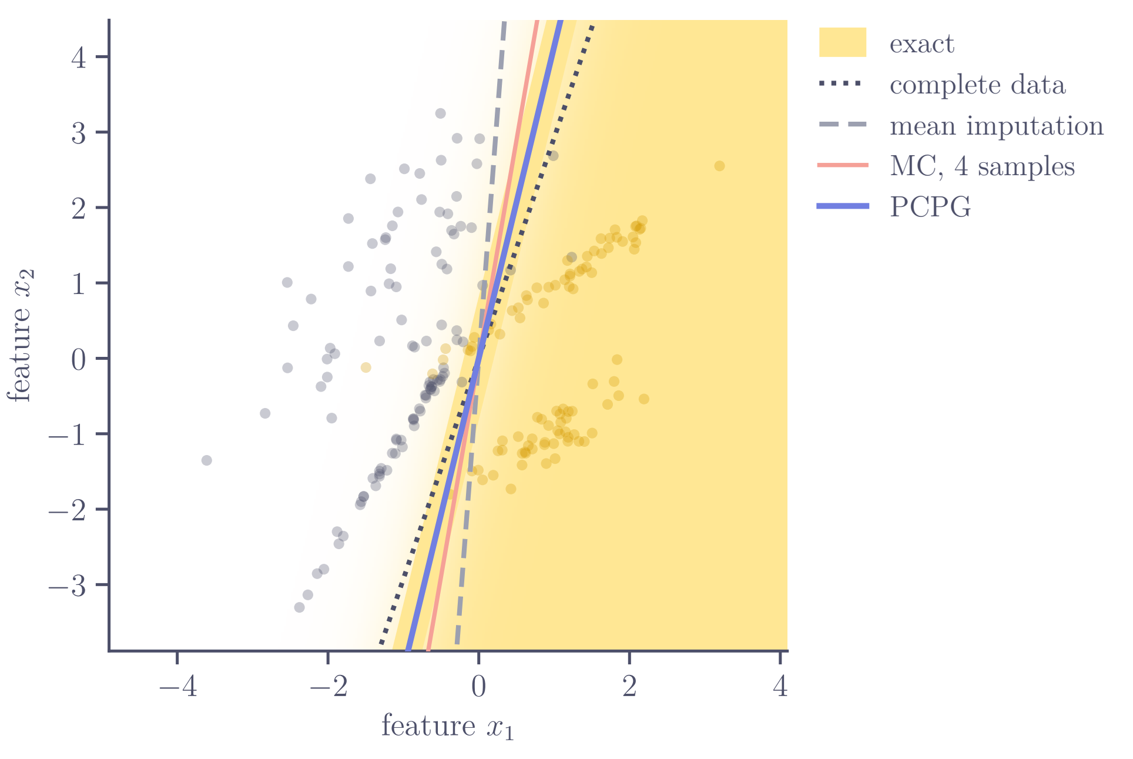
The paths of w during training

The loss during training

Training then changes w a little in the direction that lowers the loss, and starts again with step 1 at the new w, again and again from $w = 0$ (dots: every 20 steps, ring: the w of steps 1 and 2). PCPG's predictions are exact, so it walks the same path as exact training, to w^*. MC's are random, and $-\log$ of a random prediction is too large on average, the more the longer w. So MC with 4 samples stops early (cross), with 320 samples it gets close. The loss during training, minus its lowest value. PCPG keeps going down like exact training, MC with 4 samples stops far above.

RESULT

Where training ends



The boundaries at the end of training. PCPG ends on the exact one (yellow), close to the classifier trained on complete data, which saw all features. MC with 4 samples ends with a shorter w ($\|w\| = 2.62$ against 3.31): a similar boundary, but a less sure classifier. Mean imputation, which fills in the average of x_2, turns the boundary by 9°.